

BSSF Balance Enquiry REST Interface 3.9

INTERFACE DESCRIPTION

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1 Introduction

Target Groups

The target group for this document is:

— Solution Integrators

1.1 Transfer Mechanism

The transfer mechanism used is HTTPS. Both requests and responses use headers to send information about the message.

The access port is configurable. Clients that use this interface should support a configurable port.

1.2 Authentication and Authorization

Authentication is controlled by the inclusion of a Java Web Token (JWT) in the authorization header of the HTTP message.

1.3 Compatibility Mechanism

The following version mechanisms are used:

1.3.1 HTTP Content Negotiation

HTTP content negotiation uses the profile parameter in the HTTP Accept header to request a certain version of the resource representation. The profile value indicates the JSON schema and version used for the resource representation. If the profile parameter is not passed in the HTTP Accept header, the server returns the latest version of the resource representation.

For operations supporting error response version 1, if error response version is not specified it is assumed to be error response version 1.

1.3.2 Product Customization

Elements and data types that are specific for product customization are marked with PC in the section describing the element or data type. Functions and characteristics of a product customization is not part of the standard delivery. Contact your local Ericsson office for more information regarding ordering and compatibility.



All elements of a PC message are also product customizations when inside such a message. If an element or data type is marked PC within an otherwise standard record, is noted as such in the relevant chapters.

In order to improve readability the following applies in the rest of the document:

- The following applies to functions marked with (PC). The PC-tag is placed in the description.

1.3.3 Backward Compatibility

Json objects name and/or their respective content stated in this document may be changed without any further notice.

Backward compatibility within a major version is only kept for the messages sent over the network.



2 Syntax Notation

This interface uses a generic syntax that is described in the following sections. For a description of the actual values, refer to Resources.

2.1 Resource Identifier

The resource identifier is expressed by a URI with the following format:

```
{serverRoot}/{api}/{resourcePath}
```

```
{scheme:}://{host}[:{port}]/[{deploymentComponent}]{apiName}/{apiVersion}/{resourcePath}
```

Table 1 Server root parameter

Parameter	Description
serverRoot	Is implementation specific
api	Is the name and version of the API.

Table 2 API and resource path parameters

Parameter	Description
scheme	The Application protocol, for example HTTPS.
host	The network address to access the interface and its connected service.
port	The TCP/IP port used for access. If not specified, default ports for the applications are assumed.
deploymentComponent	The name of the deployment component providing the resource.
apiName	The name of the interface.
apiVersion	The version to use
resourcePath	The resource name and a unique ID.

The full URI must be encoded according to RFC3986 before usage.

2.2 HTTP Headers

The interface uses standard HTTP headers. In addition, custom HTTP headers are used with the following syntax:



ERICSSON.<name>:<value>

Accept Header Profile Value

The server accepts the following formats for profile values:

Accept: application/json; application/merge-patch+json;
profile=<profile-URI> <profile-URI>

If no Accept header field is present, then it is assumed that the client accepts all media types.

2.3 Resource Representation Profile

The profile link relation type indicates that a resource representation follows one or more profiles. For more information, refer to [RFC 6906](#).

The profile is identified by a URI with the following syntax:

profile="http://ericsson.com/<domain>.<interface>.<resource>.<operationAndDirection>.<version>.json#"

Table 3 REST Resource Representation Parameters

Parameter	Description
domain	Fixed value: bss
interface	Is the interface name.
resource	Is the resource name.
operationAndDirection	Is the CRUD operation with its direction request and response.
version	Major version of the resource representation.

3 Service

3.1 BSSF Balance Enquiry

Please read the service documentation available to understand how to use this interface in connection with the service.

3.1.1 Description

Provides balance statements for all accounts associated with a customer, contract or product.

3.2 Service Registry

The Service Registry is a central repository of available services. Their locations are used by clients to dynamically find the service providers it needs. A service provider registers its services in the registry. Service consumers look up services and service locations.

The services accessed through the interface are the following:

Table 4 Services Exposed in the Service Registry

Service Name	Service Type	Version	Environment	Available Capabilities	Optional Data
BSSF_Balance_Enquiry	REST	3	Prod		
BSSF_Balance_Enquiry	REST	3	Prod		secured=true

Note: This chapter is only applicable for Virtualized deployments.





4 Resources

The interface offers access to different resources. Each resource has its own access path and parameters which are described in the following sections.

The referenced representations are available in a ZIP file at the following location:

<edw:/alex?FN=BSSFBalanceEnquiry v3 9 schemaFile.zip>

4.1 Balance Top Up Details

The BalanceTopUp resource represents information for balance top-up.

4.1.1 Read Balance top Up Details

Read balance top-up information.

4.1.1.1 Operations

The following HTTP methods and URIs are applicable to perform the operation:

— GET {scheme:}://{host}[:{port}]/{deploymentComponent}/bssfBalanceEnquiry/v1/balanceTopUpDetails

Query Parameters

The following table describes the query parameters for the operation:

Table 5 Query Parameters

Name	Type	Description
communicationId	string	Value of Communication ID. Additional Description: Communication ID value. Maxlength: 512
communicationIdType	string	Type of Communication ID. Additional Description: Communication ID type. Maxlength: 32



Name	Type	Description
relatedPartyReferredType	string	CUSTOMER or PARTY_ROLE. When no value is provided, "CUSTOMER" is adopted as a default referred type for the related party. Additional Description: Referred Type. Maxlength: 128
customerId	string	Customer ID - e.g. 197696480485144. Additional Description: Unique identifier for the party role. Deprecated Maxlength: 256 Pattern: ^(?=\s*\S).*\$
relatedPartyId	string	Customer ID or Party Role ID - e.g. 197696480485144. Additional Description: Unique identifier of the party or party role. Maxlength: 256 Pattern: ^(?=\s*\S).*\$
contractId	string	Contract ID - e.g. 197696480485145. Additional Description: Unique identifier of the contract. Maxlength: 256 Pattern: ^(?=\s*\S).*\$



Name	Type	Description
resourceFacingServiceSpecId	string	Resource Facing Service Specification ID. Additional Description: Unique identifier of the specification. Maxlength: 256 Pattern: ^(?=\s*\S).*\$
customerExternalId	string	Customer External Identifier. Additional Description: External identifier for the party role. Deprecated Maxlength: 256 Pattern: ^(?=\s*\S).*\$
relatedPartyExternalId	string	External Identifier of a Customer or a Party Role. Additional Description: External identifier of the party or party role. Maxlength: 256 Pattern: ^(?=\s*\S).*\$



Name	Type	Description
contractExternalId	string	Contract External ID. Additional Description: External identifier of the contract for interaction in external system. Maxlength: 256 Pattern: ^(?=\s*\S).*\$
resourceFacingServiceSpecExternalId	string	Resource Facing Service Specification External ID. Additional Description: Identifier of the specification in an external system. Maxlength: 256 Pattern: ^(?=\s*\S).*\$

4.1.1.2

HTTP Headers

The following standard and custom HTTP headers are supported.

Request

HTTP headers available for the request are as follows:

Table 6 Request HTTP Headers

Header Field Name	Header Field Value	Mandatory/Optional
Accept		Optional



Header Field Name	Header Field Value	Mandatory/Optional
	A non-empty subset of the following: • application/json;profile="http://ericsson.com/bss.bssfBalanceEnquiry.balanceTopUpDetails.readResponse.3.json#";charset=utf-8 • application/json;profile="http://ericsson.com/bss.bssfBalanceEnquiry.balanceTopUpDetails.readResponse.3.json#" • application/json;profile="http://ericsson.com/bss.errorMessage.response.1.json#";charset=utf-8 • application/json;profile="http://ericsson.com/bss.errorMessage.response.1.json#" • application/json;profile="http://ericsson.com/bss.errorMessage.response.2.json#";charset=utf-8 • application/json;profile="http://ericsson.com/bss.errorMessage.response.2.json#" • application/json;charset=utf-8 • application/json	-
Authorization		Optional by default
	Bearer <token>	Mandatory If Authentication and Authorization is active then this parameter is mandatory.



Table 7 Customized Request HTTP Headers

Header Field Name	Header Field Value	Mandatory/Optional	Description
ERICSSON.Generic-Interface-Parameters	BASE64 encoded and GZIP Compressed bss.genericInterfaceParameters.request.4.json	Optional	Generic Interface Parameters.

Response

HTTP headers available for the response are as follows:

Table 8 Response HTTP Headers

Header Field Name	Header Field Value	Mandatory/Optional
Content-Type		Mandatory



Header Field Name	Header Field Value	Mandatory/Optional
	For Status-Code: 200, one of the following: • application/json;profile="http://ericsson.com/bss.bssfBalanceEnquiry.balanceTopUpDetails.readResponse.3.json#";charset=utf-8 • application/json;profile="http://ericsson.com/bss.bssfBalanceEnquiry.balanceTopUpDetails.readResponse.3.json#" For Status-Code: 4XX,5XX, one of the following: • application/json;profile="http://ericsson.com/bss.errorMessage.response.1.json#";charset=utf-8 • application/json;profile="http://ericsson.com/bss.errorMessage.response.1.json#" • application/json;profile="http://ericsson.com/bss.errorMessage.response.2.json#";charset=utf-8 • application/json;profile="http://ericsson.com/bss.errorMessage.response.2.json#"	-



Header Field Name	Header Field Value	Mandatory/Optional
Retry-After	String	Indicates an expected wait time before making a retry, as delay in seconds or HTTP-date format. Example value 'Fri, 31 Dec 1999 23:59:59 GMT' as in http-date format or '60' as delay in seconds. Optional Status Code: 503 Pattern: ^(([0-9]{1,8}) ((Mon Tue Wed Thu Fri Sat Sun), ([0-3][0-9]) (Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec) ([0-9]{4}) ([01][0-9] 2[0-3])(:[0-5][0-9]){2} GMT))\$
WWW-Authenticate		Optional Status Code: 401
	<challenge>	-
STANDARD HEADER		



Table 9 Customized Response HTTP Headers

Header Field Name	Header Field Value	Mandatory/Optional	Description
ERICSSON.Generic-Interface-Parameters	BASE64 encoded and GZIP compressed bss.genericInterfaceParameters.response.4.json	Optional	Generic Interface Parameters

GURID HEADER CUSTOM HEADER

4.1.1.3 Resource Representation - Response

The following images and tables describe the elements and their structure contained in the resource representation response.

4.1.1.3.1 bss.bssfBalanceEnquiry.balanceTopUpDetails Read Response Version 3

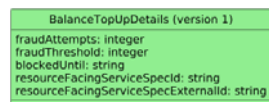


Figure 1 bss.bssfBalanceEnquiry.balanceTopUpDetails.readResponse.3 UML representation

Applicable profile URI for the representation:

<http://ericsson.com/bss.bssfBalanceEnquiry.balanceTopUpDetails.readResponse.3.json#>

The representation is provided as JSON objects with the following attributes:

Table 10 BalanceTopUpDetails (version 1)

Name	Type	Description
BalanceTopUpDetails (version 1)	object	Balance topup details.



Name		Type	Description
	fraudAttempts	integer	Number of detected fraud attempts. The numeric value must be at least -2147483648. The numeric value must not exceed 2147483647.
	fraudThreshold	integer	Fraud attempts Threshold. The numeric value must be at least -2147483648. The numeric value must not exceed 2147483647.
	blockedUntil	string	The balance top up service is blocked up until this point in time. Format: date-time Pattern: <code>^\d{4}-(((0[13578] 1[02])-(0[1-9] 12)[0-9] 3[01])) (02-(0[1-9] 12)[0-9]) ((0[469] 11)-(0[1-9] 12)[0-9] 30)))T((([01][0-9] 2[0-3]):[0-5][0-9]:[0-5][0-9]\.[0-9]{3}) (24:00:00.000))(Z \\+((0[0-9] 1[0-3]):[0-5][0-9] 14:00) -((0[0-9] 1[0-1]):[0-5][0-9] 12:00))\$</code>
	resourceFacingServiceSpecId	string	A unique reference to the Resource Facing Service Specification. Pattern: <code>^(?=\s*\S).*</code> Max length: 256
	resourceFacingServiceSpecExternalId	string	An external reference to the Resource Facing Service Specification. Pattern: <code>^(?=\s*\S).*</code> Max length: 256



4.2 Bucket

The Bucket resource represents product buckets or billing account buckets or settlement account buckets.

4.2.1 Read Balance

Read balances for product buckets or billing account buckets or settlement account buckets.

4.2.1.1 Operations

The following HTTP methods and URIs are applicable to perform the operation:

— GET {scheme:}://{host}[:{port}]/{deploymentComponent}/bssfBalanceEnquiry/v1/balance

Query Parameters

The following table describes the query parameters for the operation:

Table 11 Query Parameters

Name	Type	Description
communicationId	string	Value of Communication ID. Additional Description: Communication ID value. Maxlength: 512
communicationIdType	string	Type of Communication ID. Additional Description: Communication ID type. Maxlength: 32



Name	Type	Description
relatedPartyReferredType	string	CUSTOMER or PARTY_ROLE. When no value is provided, "CUSTOMER" is adopted as a default referred type for the related party. Additional Description: Referred Type. Maxlength: 128
customerId	string	Customer ID - e.g. 197696480485144. Additional Description: Unique identifier for the party role. Deprecated Maxlength: 256 Pattern: ^(?=\s*\S).*\$
relatedPartyId	string	Customer ID or Party Role ID - e.g. 197696480485144. Additional Description: Unique identifier of the party or party role. Maxlength: 256 Pattern: ^(?=\s*\S).*\$
contractId	string	Contract ID - e.g. 197696480485145. Additional Description: Unique identifier of the contract. Maxlength: 256 Pattern: ^(?=\s*\S).*\$



Name	Type	Description
productId	string	Product ID - e.g. P_50ExtraSMSPerWeek. Additional Description: Unique identifier of a product. Maxlength: 256 Pattern: ^(?=\s*\S).*\$
bucketSpecId	string	The Bucket Spec Id for a customer Bucket. Additional Description: Unique identifier of the specification. Maxlength: 256 Pattern: ^(?=\s*\S).*\$
customerExternalId	string	Customer External Identifier. Additional Description: External identifier for the party role. Deprecated Maxlength: 256 Pattern: ^(?=\s*\S).*\$
relatedPartyExternalId	string	External Identifier of a Customer or a Party Role. Additional Description: External identifier of the party or party role. Maxlength: 256 Pattern: ^(?=\s*\S).*\$



Name	Type	Description
contractExternalId	string	Contract External ID. Additional Description: External identifier of the contract for interaction in external system. Maxlength: 256 Pattern: ^(?:=\s*\S).*
productExternalId	string	Product External ID. Additional Description: Identifier of the product in an external system. Maxlength: 256 Pattern: ^(?:=\s*\S).*
bucketSpecExternalId	string	External ID of the Bucket Spec. Additional Description: Identifier of the specification in an external system. Maxlength: 256 Pattern: ^(?:=\s*\S).*
balanceType	string	Balance type. Additional Description: Type of balance. Possible enum values: ALL BILLING_ACCOUNT PRODUCT SETTLEMENT_ACCOUNT all ⁽¹⁾ product ⁽²⁾ billingAccount ⁽³⁾



Name	Type	Description
includeReservations	boolean	Include information on reserved amount on Billing Account Bucket or Product Bucket. The values "true", "True", "false", and "False" can be used. Additional Description: Boolean data type.
sharedBalanceInAggregation	boolean	Include the balance from the provider in the consuming contract, during aggregation. The values "true", "True", "false", and "False" can be used. Additional Description: Boolean data type.
balanceTimeStamp	string	Balance time stamp Additional Description: balance timestamp. Format: date-time Pattern: <code>^\d{4}-(((0[13578] 1[02])-(0[1-9] 12)[0-9] 3[01])) (02-(0[1-9] 12)[0-9]) ((0[469] 11)-(0[1-9] 12)[0-9] 30)))T((([01][0-9] 2[0-3]):[0-5][0-9]:[0-5][0-9]\.[0-9]{3}) (24:00:00.000))(Z \\+((0[0-9] 1[0-3]):[0-5][0-9] 14:00) -((0[0-9] 1[0-1]):[0-5][0-9] 12:00)))\$</code>

(1) Deprecated

(2) Deprecated

(3) Deprecated

4.2.1.2

HTTP Headers

The following standard and custom HTTP headers are supported.



Request

HTTP headers available for the request are as follows:

Table 12 Request HTTP Headers

Header Field Name	Header Field Value	Mandatory/Optional
Accept		Optional
	A non-empty subset of the following: • application/json;profile="http://ericsson.com/bss.bssfBalanceEnquiry.bucket.readResponse.3.json#";charset=utf-8 • application/json;profile="http://ericsson.com/bss.bssfBalanceEnquiry.bucket.readResponse.3.json#" • application/json;profile="http://ericsson.com/bss.errorMessage.response.1.json#";charset=utf-8 • application/json;profile="http://ericsson.com/bss.errorMessage.response.1.json#" • application/json;profile="http://ericsson.com/bss.errorMessage.response.2.json#";charset=utf-8 • application/json;profile="http://ericsson.com/bss.errorMessage.response.2.json#" • application/json;charset=utf-8 • application/json	-
Authorization		Optional by default
	Bearer <token>	Mandatory If Authentication and Authorization is active then this parameter is mandatory.



Table 13 Customized Request HTTP Headers

Header Field Name	Header Field Value	Mandatory/Optional	Description
ERICSSON.Generic-Interface-Parameters	BASE64 encoded and GZIP Compressed bss.genericInterfaceParameters.request.4.json	Optional	Generic Interface Parameters.

Response

HTTP headers available for the response are as follows:

Table 14 Response HTTP Headers

Header Field Name	Header Field Value	Mandatory/Optional
Content-Type		Mandatory
	For Status-Code: 200, one of the following: • application/json;profile="http://ericsson.com/bss.bssfBalanceEnquiry.bucket.readResponse.3.json#";charset=utf-8 • application/json;profile="http://ericsson.com/bss.bssfBalanceEnquiry.bucket.readResponse.3.json#" For Status-Code: 4XX,5XX, one of the following: • application/json;profile="http://ericsson.com/bss.errorMessage.response.1.json#";charset=utf-8 • application/json;profile="http://ericsson.com/bss.errorMessage.response.1.json#" • application/json;profile="http://ericsson.com/bss.errorMessage.response.2.json#";charset=utf-8 • application/json;profile="http://ericsson.com/bss.errorMessage.response.2.json#"	-



Header Field Name	Header Field Value	Mandatory/Optional
Retry-After	String	Indicates an expected wait time before making a retry, as delay in seconds or HTTP-date format. Example value 'Fri, 31 Dec 1999 23:59:59 GMT' as in http-date format or '60' as delay in seconds. Optional Status Code: 503 Pattern: ^(([0-9]{1,8}) ((Mon Tue Wed Thu Fri Sat Sun), ([0-3][0-9]) (Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec) ([0-9]{4}) ([01][0-9] 2[0-3])(:[0-5][0-9]){2} GMT))\$
WWW-Authenticate		Optional Status Code: 401
	<challenge>	-
STANDARD HEADER		

Table 15 Customized Response HTTP Headers

Header Field Name	Header Field Value	Mandatory/Optional	Description
ERICSSON.Generic-Interface-Parameters	BASE64 encoded and GZIP compressed bss.genericInterfaceParameters.response.4.json	Optional	Generic Interface Parameters

GURID HEADER CUSTOM HEADER

4.2.1.3 Resource Representation - Response

The following images and tables describe the elements and their structure contained in the resource representation response.

4.2.1.3.1 bss.bssfBalanceEnquiry.bucket Read Response Version 3

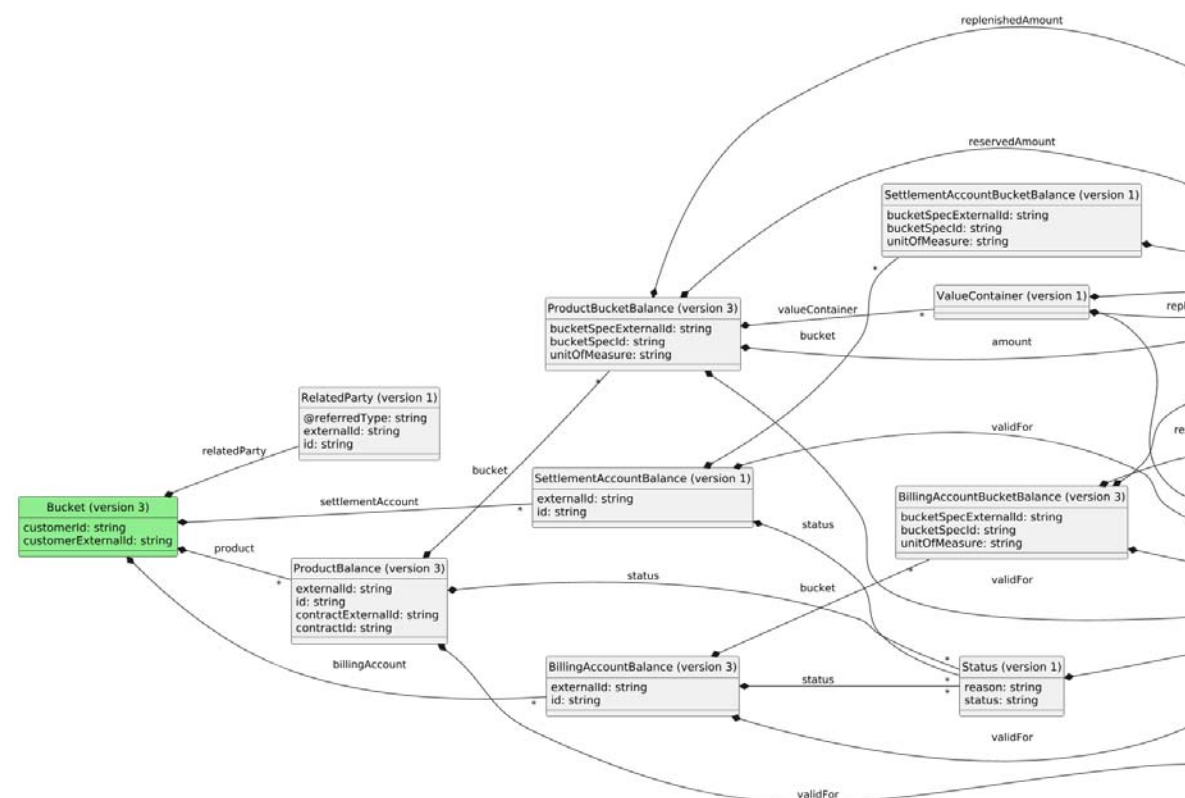


Figure 2 `bss.bssfBalanceEnquiry.bucket.readResponse.3` UML representation

Applicable profile URI for the representation:



`http://ericsson.com/bss.bssfBalanceEnquiry.bucket.readResponse.3.json#`

The representation is provided as JSON objects with the following attributes:

Table 16 Bucket (version 3)

Name		Type	Description
Bucket (version 3)		object	Bucket.
	customerId	string	Represents the customer ID. Pattern: <code>^(?=\s*\S).*</code> Max length: 256 Deprecated
	customerExternalId	string	Represents the external customer ID. Pattern: <code>^(?=\s*\S).*</code> Max length: 256 Deprecated
	relatedParty	RelatedParty (version 1)	Reference to the related Party Role
	settlementAccount	array of SettlementAccountBalance (version 1)	Settlement accounts.
	billingAccount	array of BillingAccountBalance (version 3)	Billing accounts.
	product	array of ProductBalance (version 3)	Products.

RelatedParty (version 1)

Table 17 RelatedParty (version 1)

Name	Type	Description
RelatedParty (version 1)	object	Related Party or Party Role



Name		Type	Description
	@referredType	string	Referred Type. Max length: 128
	externalId	string	External identifier of the party or party role. Pattern: ^(?=\s*\S).*\$ Max length: 256
	id	string	Unique identifier of the party or party role. Pattern: ^(?=\s*\S).*\$ Max length: 256

SettlementAccountBalance (version 1)

Table 18 SettlementAccountBalance (version 1)

Name		Type	Description
SettlementAccountBalance (version 1)		object	A settlement account balance structure.
	externalId	string	Unique identifier of the settlement account populated by external system. Pattern: ^(?=\s*\S).*\$ Max length: 256
	id	string	Unique identifier of the settlement account. Pattern: ^(?=\s*\S).*\$ Max length: 256
	validFor	bss.cmn.TimePeriod (version 1)	The time period that the settlement account is valid for.
	status	array of bss.cmn.Status (version 1)	The status of the settlement account, for example active.
	bucket	array of SettlementAccountBucketBalance (version 1)	Buckets.



bss.cmn.TimePeriod (version 1)

Table 19 bss.cmn.TimePeriod (version 1)

Name		Type	Description
bss.cmn.TimePeriod (version 1)		object	StartDateTime must be earlier than EndDateTime.
	startDateTime	string	Start dateTime of the time period. The value is inclusive, including the starting point of the interval. The pattern is based on the definition found at http://www.w3.org/TR/xmlschema-2/#dateTime. Format: date-time Pattern: <code>^\d{4}-(((0[13578] 1[02])-(0[1-9] 12)[0-9] 3[01])) (02-(0[1-9] 12)[0-9]) ((0[469] 11)-(0[1-9] 12)[0-9] 30)))T((([01][0-9] 2[0-3]):[0-5][0-9]:[0-5][0-9]\.[0-9]{3}) (24:00:00.000))(Z \\+((0[0-9] 1[0-3]):[0-5][0-9] 14:00) -((0[0-9] 1[0-1]):[0-5][0-9] 12:00))\$</code>
	endDateTime	string	End dateTime of the time period. The value is exclusive, excluding the ending point of the interval. The pattern is based on the definition found at http://www.w3.org/TR/xmlschema-2/#dateTime. Format: date-time Pattern: <code>^\d{4}-(((0[13578] 1[02])-(0[1-9] 12)[0-9] 3[01])) (02-(0[1-9] 12)[0-9]) ((0[469] 11)-(0[1-9] 12)[0-9] 30)))T((([01][0-9] 2[0-3]):[0-5][0-9]:[0-5][0-9]\.[0-9]{3}) (24:00:00.000))(Z \\+((0[0-9] 1[0-3]):[0-5][0-9] 14:00) -((0[0-9] 1[0-1]):[0-5][0-9] 12:00))\$</code>



bss.cmn.Status (version 1)

Table 20 bss.cmn.Status (version 1)

Name		Type	Description
bss.cmn.Status (version 1)		object	Structure of one status version element.
	reason	string	The reason for the new status version. Max length: 256
	status	string	The status for the given time period. Max length: 256
	validFor	bss.cmn.TimePeriod (version 1)	Time period where the status is valid.

SettlementAccountBucketBalance (version 1)

Table 21 SettlementAccountBucketBalance (version 1)

Name		Type	Description
SettlementAccountBucketBalance (version 1)		object	Settlement account bucket balance.
	bucketSpecExternalId	string	Uniquely identifying the bucket using the external specification id. Pattern: ^(?=\s*\S).* Max length: 256
	bucketSpecId	string	Uniquely identifying the bucket using the specification id. Pattern: ^(?=\s*\S).* Max length: 256
	amount	bss.cmn.Decimal_Sensitive (version 1)	The balance of the Settlement Account Bucket.
	unitOfMeasure	string	Unit of measure for amounts. Max length: 32



bss.cmn.Decimal_Sensitive (version 1)

Table 22 bss.cmn.Decimal_Sensitive (version 1)

Name		Type	Description
bss.cmn.Decimal_Sensitive (version 1)		object	Amount representation.
	number	integer	Carrying value. The numeric value must be at least -9223372036854775807. The numeric value must not exceed 9223372036854775806.
	decimalPlaces	integer	Number of decimals in the value. The numeric value must be at least 0. The numeric value must not exceed 6.

BillingAccountBalance (version 3)

Table 23 BillingAccountBalance (version 3)

Name	Type	Description
BillingAccountBalance (version 3)	object	A billing account balance structure.



Name		Type	Description
	externalId	string	Unique identifier of the billing account populated by external system. Pattern: ^(?=\s*\S).* Max length: 256
	id	string	Unique identifier of the billing account. Pattern: ^(?=\s*\S).* Max length: 256
	validFor	bss.cmn.TimePeriod (version 1)	The time period that the billing account is valid for.
	status	array of bss.cmn.Status (version 1)	The status of the billing account, for example active.
	bucket	array of BillingAccountBucketBalance (version 3)	Buckets.

BillingAccountBucketBalance (version 3)

Table 24 BillingAccountBucketBalance (version 3)

Name	Type	Description
BillingAccountBucketBalance (version 3)	object	Billing account bucket balance.



Name		Type	Description
	bucketSpecExternalId	string	Uniquely identifying the bucket using the external specification id. Pattern: ^(?=\s*\S).*\$ Max length: 256
	bucketSpecId	string	Uniquely identifying the bucket using the specification id. Pattern: ^(?=\s*\S).*\$ Max length: 256
	amount	bss.cmn.Decimal_Sensitive (version 1)	The balance of the Billing Account Bucket.
	reservedAmount	bss.cmn.Decimal_Sensitive (version 1)	The reserved amount on the Billing Account Bucket.
	validFor	bss.cmn.TimePeriod (version 1)	Validity of the bucket.
	unitOfMeasure	string	Unit of measure for amounts. Max length: 32

ProductBalance (version 3)

Table 25 ProductBalance (version 3)

Name	Type	Description
ProductBalance (version 3)	object	Product Balance.



Name		Type	Description
	externalId	string	Unique identifier of the product populated by external system. Pattern: ^(?=\s*\S).*\$ Max length: 256
	id	string	Unique identifier of the product. Pattern: ^(?=\s*\S).*\$ Max length: 256
	contractExternalId	string	External identifier of the contract where the product belongs. Pattern: ^(?=\s*\S).*\$ Max length: 256
	contractId	string	Unique identifier of the contract where the product belongs. Pattern: ^(?=\s*\S).*\$ Max length: 256
	validFor	bss.cmn.TimePeriod (version 1)	The time period that the product is valid for.
	status	array of bss.cmn.Status (version 1)	The status of the product.
	bucket	array of ProductBucketBalance (version 3)	The product buckets.

ProductBucketBalance (version 3)

Table 26 ProductBucketBalance (version 3)

Name	Type	Description
ProductBucketBalance (version 3)	object	Product Bucket balance.



Name		Type	Description
	bucketSpecExternalId	string	Uniquely identifying the bucket using the external specification id. Pattern: ^(?=\s*\S).*\$ Max length: 256
	bucketSpecId	string	Uniquely identifying the bucket using the specification id. Pattern: ^(?=\s*\S).*\$ Max length: 256
	amount	bss.cmn.Decimal_Sensitive (version 1)	The amount of the bucket.
	replenishedAmount	bss.cmn.Decimal_Sensitive (version 1)	The replenished amount of the bucket. PC:27123
	reservedAmount	bss.cmn.Decimal_Sensitive (version 1)	The reserved amount on the Bucket.
	validFor	bss.cmn.TimePeriod (version 1)	Validity of the bucket.
	unitOfMeasure	string	Unit of measure for amounts. Max length: 32
	valueContainer	array of ValueContainer (version 1)	Value containers.

ValueContainer (version 1)

Table 27 ValueContainer (version 1)

Name		Type	Description
ValueContainer (version 1)		object	Value Container.
	amount	bss.cmn.Decimal_Sensitive (version 1)	The amount of this value container.
	replenishedAmount	bss.cmn.Decimal_Sensitive (version 1)	The replenished amount of the value container. PC:27123
	validFor	bss.cmn.TimePeriod (version 1)	The validity period of the value container.



5 Generic Interface Parameters

The following sub-sections describe the elements and their structure in the request and response for Generic Interface Parameters.

5.1 Generic Interface Parameters Request

5.1.1 bss.genericInterfaceParameters Request Version 4

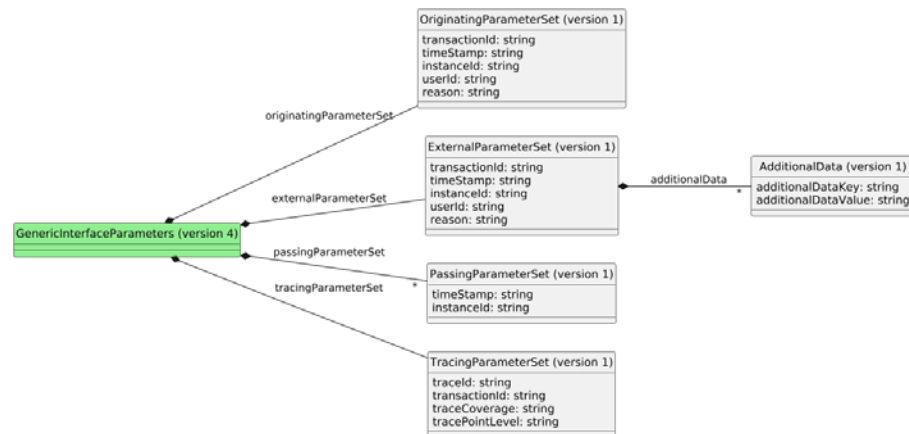


Figure 3 bss.genericInterfaceParameters.request.4 UML representation

Applicable profile URI for the representation:

<http://ericsson.com/bss.genericInterfaceParameters.request.4.json#>

The representation is provided as JSON objects with the following attributes:

Table 28 bss.cmn.GenericInterfaceParameters (version 4)

Name	Type	Description
bss.cmn.GenericInterfaceParameters (version 4)	object	The Generic Interface Parameters (GIP) is used for transaction traceability.



Name		Type	Description
	originatingParameterSet	bss.cmn.OriginatingParameterSet (version 1)	Transaction Originating Parameter Set. Identification of the transaction. Identification of the instance that created the ops.
	externalParameterSet	bss.cmn.ExternalParameterSet (version 1)	Information received from the external system that initiated the transaction.
	passingParameterSet	array of bss.cmn.PassingParameterSet (version 1)	Transaction Passing Parameter Set.
	tracingParameterSet	bss.cmn.TracingParameterSet (version 1)	Tracing Parameter Set.

5.1.1.1

bss.cmn.OriginatingParameterSet (version 1)

Table 29 bss.cmn.OriginatingParameterSet (version 1)

Name	Type	Description
bss.cmn.OriginatingParameterSet (version 1)	object	Transaction Originating Parameter Set. Identification of the transaction. Identification of the instance that created the ops.



Name		Type	Description
	transactionId	string	Unique transaction identifier. Pattern: <code>^[a-fA-F0-9]{8}-?[a-fA-F0-9]{4}-?[a-fA-F0-9]{4}-?[a-fA-F0-9]{4}-?[a-fA-F0-9]{12}\$</code>
	timeStamp	string	The time stamp when the ops was created. Shall adhere to W3 date-time format. Format: date-time Pattern: <code>^\d{4}-((0[13578] 1[02])-(0[1-9] 12)[0-9] 3[01])) ((02-(0[1-9] 12)[0-9]) ((0[469] 11)-(0[1-9] 12)[0-9] 30)))T((0[1-9] 2[0-3]):[0-5][0-9]:[0-5][0-9]\.[0-9]{3}) (24:00:00.000))(Z (\+((0[0-9] 1[0-3]):[0-5][0-9] 14:00) -((0[0-9] 1[0-1]):[0-5][0-9] 12:00)))\$</code>
	instanceId	string	Identification of the instance that created the ops. Max length: 256
	userId	string	User Identification. Provided at login to the instance creating the ops. Max length: 256
	reason	string	Reason for the transaction. Max length: 256

5.1.1.2

bss.cmn.ExternalParameterSet (version 1)

Table 30 bss.cmn.ExternalParameterSet (version 1)

Name	Type	Description
bss.cmn.ExternalParameterSet (version 1)	object	Information received from the external system that initiated the transaction.



Name		Type	Description
	transactionId	string	Transaction Identifier received from external system. Max length: 256
	timeStamp	string	Transaction time stamp received from external system. Shall adhere to W3 date-time format. Format: date-time Pattern: <code>^\d{4}-(((0[13578] 1[02])-(0[1-9] 12)[0-9] 3[01])) (02-(0[1-9] 12)[0-9]) ((0[469] 11)-(0[1-9] 12)[0-9] 30)))T((([01][0-9] 2[0-3]):[0-5][0-9]:[0-5][0-9]\\.([0-9]{3} (24:00:00.000)))(Z \\+((0[0-9] 1[0-3]):[0-5][0-9] 14:00) -((0[0-9] 1[0-1]):[0-5][0-9] 12:00)))\$</code>
	instanceId	string	Identification of the instance that sent the transactionId. Max length: 256
	userId	string	User identifier received from external system. Max length: 256
	reason	string	Reason received from external system. Max length: 256
	additionalData	array of bss.cmn.AdditionalData (version 1)	Array of additional data Items from external system.

5.1.1.3

bss.cmn.AdditionalData (version 1)

Table 31 bss.cmn.AdditionalData (version 1)

Name	Type	Description
bss.cmn.AdditionalData (version 1)	object	A key-value structure data received from external system.



Name		Type	Description
	additionalDataKey	string	key of the additional data from external system. Max length: 256
	additionalDataValue	string	value correspond to the key. Max length: 256

5.1.1.4

bss.cmn.PassingParameterSet (version 1)

Table 32 bss.cmn.PassingParameterSet (version 1)

Name		Type	Description
bss.cmn.PassingParameterSet (version 1)		object	Identification of an instance where the transaction has passed.
	timeStamp	string	Time stamp at which the ppsItem was created. Shall adhere to W3 date-time format. Format: date-time Pattern: <code>^\d{4}-(((0[13578] 1[02])-(0[1-9] [12][0-9] 3[01])) (02-(0[1-9] [12][0-9])) ((0[469] 11)-(0[1-9] [12][0-9] 3[0])))T(((01[0-9] 2[0-3]):[0-5][0-9]:[0-5][0-9]\.[0-9]{3}) (24:00:00.000))(Z \\+((0[0-9] 1[0-3]):[0-5][0-9] 14:00) -((0[0-9] 1[0-1]):[0-5][0-9] 12:00))\$</code>
	instanceId	string	Identification of the entity that created the ppsItem. Max length: 256

5.1.1.5

bss.cmn.TracingParameterSet (version 1)

Table 33 bss.cmn.TracingParameterSet (version 1)

Name		Type	Description
bss.cmn.TracingParameterSet (version 1)		object	Tracing Parameter Set.



Name		Type	Description
	traceId	string	The unique id for the trace job. Pattern: <code>^[a-zA-F0-9]{8}-?[a-zA-F0-9]{4}-?[a-zA-F0-9]{4}-?[a-zA-F0-9]{4}-?[a-zA-F0-9]{12}\$</code>
	transactionId	string	The unique id for the transaction. Pattern: <code>^[a-zA-F0-9]{8}-?[a-zA-F0-9]{4}-?[a-zA-F0-9]{4}-?[a-zA-F0-9]{4}-?[a-zA-F0-9]{12}\$</code>
	traceCoverage	string	The trace coverage indicator. Local or Global. Max length: 256
	tracePointLevel	string	The level of trace points to include in the trace. 0 means External trace points, 1 means External and internal trace points. Max length: 256

5.2 Generic Interface Parameters Response

5.2.1 bss.genericInterfaceParameters Response Version 4

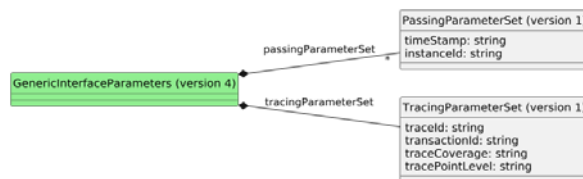


Figure 4 bss.genericInterfaceParameters.response.4 UML representation

Applicable profile URI for the representation:

`http://ericsson.com/bss.genericInterfaceParameters.response.4.json#`

The representation is provided as JSON objects with the following attributes:



Table 34 bss.cmn.GenericInterfaceParameters (version 4)

Name		Type	Description
bss.cmn.GenericInterfaceParameters (version 4)		object	The Generic Interface Parameters (GIP) is used for transaction traceability.
	passingParameterSet	array of bss.cmn.PassingParameterSet (version 1)	Transaction Passing Parameter Set.
	tracingParameterSet	bss.cmn.TracingParameterSet (version 1)	Tracing Parameter Set.

5.2.1.1

bss.cmn.PassingParameterSet (version 1)

Table 35 bss.cmn.PassingParameterSet (version 1)

Name		Type	Description
bss.cmn.PassingParameterSet (version 1)		object	Identification of an instance where the transaction has passed.
	timeStamp	string	Time stamp at which the ppsItem was created. Shall adhere to W3 date-time format. Format: date-time Pattern: <code>^\d{4}-(((0[13578] 1[02])-(0[1-9] [12][0-9] 3[01])) ((02-(0[1-9] [12][0-9])) ((0[469] 11)-(0[1-9] [12][0-9] 30)))T(((01[0-9] 2[0-3]):[0-5][0-9]:[0-5][0-9]\.[0-9]{3}) (24:00:00.000))(Z (\+((0[0-9] 1[0-3]):[0-5][0-9] 14:00) -(0[0-9] 1[0-1]):[0-5][0-9] 12:00)))\$</code>
	instanceId	string	Identification of the entity that created the ppsItem. Max length: 256



5.2.1.2 bss.cmn.TracingParameterSet (version 1)

Table 36 bss.cmn.TracingParameterSet (version 1)

Name		Type	Description
bss.cmn.TracingParameterSet (version 1)		object	Tracing Parameter Set.
	traceId	string	The unique id for the trace job. Pattern: <code>^[a-zA-F0-9]{8}-?[a-zA-F0-9]{4}-?[a-zA-F0-9]{4}-?[a-zA-F0-9]{4}-?[a-zA-F0-9]{12}\$</code>
	transactionId	string	The unique id for the transaction. Pattern: <code>^[a-zA-F0-9]{8}-?[a-zA-F0-9]{4}-?[a-zA-F0-9]{4}-?[a-zA-F0-9]{4}-?[a-zA-F0-9]{12}\$</code>
	traceCoverage	string	The trace coverage indicator. Local or Global. Max length: 256
	tracePointLevel	string	The level of trace points to include in the trace. 0 means External trace points, 1 means External and internal trace points. Max length: 256

6 Result Representation

Results are reported through HTTP status codes. JSON objects are provided for requests that reaches the application component. The following tables describe the elements and their structure contained in the result representation.

6.1 bss.errorMessage Response Version 1



Figure 5 bss.errorMessage.response.1 UML representation

Applicable profile URI for the representation:

`http://ericsson.com/bss.errorMessage.response.1.json#`

The representation is provided as JSON objects with the following attributes:

Table 37 bss.cmn.ErrorMessageExtended (version 1)

Name		Type	Description
bss.cmn.ErrorMessageExtended (version 1)		object	Error message.
	messages	array of bss.cmn.ErrorMessage (version 1)	Error messages.

6.1.1 bss.cmn.ErrorMessage (version 1)

Table 38 bss.cmn.ErrorMessage (version 1)

Name	Type	Description
bss.cmn.ErrorMessage (version 1)	object	Error message.

Name	Type	Description
action	string	The action. Max length: 256
code	string	The code. Max length: 256
details	string	The details. Max length: 256
message	string	The message. Max length: 256
source	string	The source. Max length: 256

6.2 bss.errorMessage Response Version 2

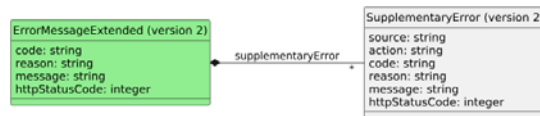


Figure 6 bss.errorMessage.response.2 UML representation

Applicable profile URI for the representation:

<http://ericsson.com/bss.errorMessage.response.2.json#>

The representation is provided as JSON objects with the following attributes:

Table 39 bss.cmn.ErrorMessageExtended (version 2)

Name	Type	Description
bss.cmn.ErrorMessageExtended (version 2)	object	Extended error message structure.



Name		Type	Description
	supplementaryError	array of bss.cmn.SupplementaryError (version 2)	Supplementary Error.
	code	string	Application related code (as defined in the API or from a common list). Max length: 256
	reason	string	Text that explains the reason for error. This can be shown to a client user. Max length: 256
	message	string	Text that provide more details and corrective actions related to the error. This can be shown to a client user. Max length: 256
	statusCode	integer	HTTP status code.

6.2.1

bss.cmn.SupplementaryError (version 2)

Table 40 bss.cmn.SupplementaryError (version 2)

Name	Type	Description
bss.cmn.SupplementaryError (version 2)	object	Supplementary error message structure. In case of partial error scenarios, this structure may also include successful responses to show what parts went ok.



Name		Type	Description
	source	string	Source of error. Max length: 256
	action	string	Action causing error. Max length: 256
	code	string	Application related code (as defined in the API or from a common list). Max length: 256
	reason	string	Text that explains the reason for error. This can be shown to a client user. Max length: 256
	message	string	Text that provide more details and corrective actions related to the error. This can be shown to a client user. Max length: 256
	statusCode	integer	HTTP status code.

6.3 Result Codes

The supported Error Codes could be found in the 'Error Codes' document.



7 Interface Version Changes

The following sections describe the changes between the interface versions.

7.1 Interface Version 1.0

Table 41 Changes

Resources	Type of Change	Description
N/A	Document created.	First version of the protocol.

7.2 Interface Version 2.0

Table 42 Changes

Resources	Type of Change	Description
N/A	JSON file name have been updated	Interface named and domain added.
externalId	Change of attribute name	Replaced by externalSpecId.
id	Change of attribute name	Replaced by specId.
compricedOf	Removal of attribute name	Removed in productbucketbalance.

7.3 Interface Version 2.1

Table 43 Changes

Resources	Type of Change	Description
BillingAccountBucketBalance	Updated response representation	Added reservedAmount attribute.
ProductBucketBalance	Updated response representation	Added reservedAmount attribute.
	Added query parameter	Added includeReservations query parameter.
	Updated the chapter "Result Representation"	Removed Status Codes.



7.4 Interface Version 3.0

Table 44 Changes

Resources	Type of Change	Description
ProductBucketBalance	Change of attribute name	Renamed specId to bucketSpecId.
ProductBucketBalance	Change of attribute name	Renamed externalSpecId to bucketSpecExternalId.
BillingAccountBucketBalance	Change of attribute name	Renamed specId to bucketSpecId.
BillingAccountBucketBalance	Change of attribute name	Renamed externalSpecId to bucketSpecExternalId.

7.5 Interface Version 3.1

Table 45 Changes

Resources	Type of Change	Description
BillingAccountBucketBalance	Change of attribute name	Corrected externalSpecId to bucketSpecExternalId.
BillingAccountBucketBalance	Change of attribute name	Corrected specId to bucketSpecId.
ProductBucketBalance	Change of attribute name	Corrected externalSpecId to bucketSpecExternalId.
ProductBucketBalance	Change of attribute name	Corrected specId to bucketSpecId.
errorMessage	Added Supplementary structure	Added SupplementaryError structure in errorMessage json.



7.6 Interface Version 3.2

Table 46 Changes

Resources	Type of Change	Description
BalanceTopUp	New resource	
Bucket	New query parameter sharedBalanceInAggregation added.	

7.7 Interface Version 3.3

Table 47 Changes

Resources	Type of Change	Description
Error message structure version 1.	Deprecated	Deprecated attributes: httpStatusCodeDescription, errorMessage, errorDetails.
Error message structure version 2.	New	New attributes: code, reason, message.

7.8 Interface Version 3.4

Table 48 Changes

Resources	Type of Change	Description
Bucket	New query parameter	Added balanceTimeStamp query parameter.

7.9 Interface Version 3.5

Table 49 Changes

Resources	Type of Change	Description
ProductBucketBalance	New attribute	ReplenishedAmount added for PC:27123.
ValueContainer	New attribute	ReplenishedAmount added for PC:27123.



7.10 Interface Version 3.6

Table 50 Changes

Resources	Type of Change	Description
bss.cmn.Status	Data type extended	Data type for status increased from 32 to 256 characters.

7.11 Interface Version 3.7

Table 51 Changes

Resources	Type of Change	Description
Bucket	Updated	Read Bucket query parameter "balanceType" is updated with possible values.
Customer	Customer is deprecated with RelatedParty.	Customer is deprecated with RelatedParty RelatedPartyReferredType is added as a query parameter, to support balance enquiry with relatedParty. CustomerId query parameter is deprecated with relatedPartyId. CustomerExternalId query parameter is deprecated with relatedPartyExternalId.
Settlement Account	Introduced Settlement Account balance enquiry	Added the possibility to enquire settlement account balance. Settlement account is associated with a party role Partner.



7.12 Interface Version 3.8

Table 52 Changes

Resources	Type of Change	Description
All	Added	Added Retry-After header for http status code 503 response. Added ERICSSON.Global-Unique-Request-Id header for all responses. Added http status code 503 for all responses.

7.13 Interface Version 3.9

Table 53 Changes

Resources	Type of Change	Description
All	Updated	Change handling of content-type for request/responses operations.





Reference List

- [1] [RFC 7231 - Hypertext Transfer Protocol \(HTTP/1.1\): Semantics and Content](#)
- [2] [RFC 7230 - Hypertext Transfer Protocol \(HTTP/1.1\): Message Syntax and Routing](#)
- [3] [RFC 6906 - The 'profile' Link Relation Type](#)
- [4] [RFC 4122 - A Universally Unique Identifier \(UUID\) URN Namespace](#)
- [5] [RFC 3986 - Uniform Resource Identifier \(URI\): Generic Syntax](#)
- [6] [RFC 2818 - HTTP Over TLS](#)
- [7] [RFC 2616 - Hypertext Transfer Protocol -- HTTP/1.1](#)